

14. The Arithmetic Theory of Algebraic Functions of One Variable in Its Relation to the Other Theories and to Algebraic Number Theory

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The following report arose from the wish to have a connecting link between the individual theories of algebraic functions; at the same time it may be regarded as a supplement to the report by Brill–Noether,¹ in which the discussion of the arithmetic theory is essentially absent.²

For discussions of the individual theories, besides Brill–Noether, the following encyclopedia articles, with the literature indicated there, come into consideration:

For the theory of Riemann: Wirtinger (II B 2), nos. 1–19;

for the theory of Weierstrass: Wirtinger (II B 2), nos. 23, 32, 33, 34;

for the algebraic-geometric theory of Brill–Noether: Wirtinger (II B 2), nos. 21–31; Berzolari (III C 4), nos. 23–38;

for the arithmetic theory of Dedekind–Weber and Hensel–Landsberg: Hensel (II C 5), [cited as Hensel];³

more generally, for the arithmetic theory of algebraic quantities: Landsberg (I C 5), and for the arithmetic theory of algebraic functions of several variables: Jung (II C 6).

For algebraic number theory compare Hilbert’s “Zahlbericht”⁴ and the lectures on number theory by Dirichlet–Dedekind [cited as Dedekind–number theory].⁵

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1. Foundations of the various theories

Riemann’s theory is the only purely transcendental one, resting on existence theorems (Dirichlet’s principle, Schwarz–Neumann methods). The integrals of algebraic functions are here primary, the algebraic functions themselves secondary. The functions on the Riemann surface are characterized by their zeros and poles, corresponding to the polygons (respectively divisors) of the arithmetic theory; whereas Weierstrass also speaks of the functions themselves together with their zeros and poles. The main problem, apart from the existence theorems, is the setting up of all functions with given poles, which is accomplished by integrals of the first and second kind; and, further, the question of the number of arbitrary constants that still enter when the points at infinity are prescribed. This latter question is answered by the Riemann–Roch theorem; corresponding to it in the arithmetic theory is the question of the dimension of a polygon class (respectively divisor class), and in the algebraic-geometric theory the question of the dimension of a complete linear system, whose answer is also provided here by the Riemann–Roch theorem. The actual construction of the functions is carried out in the arithmetic theory by means of the basis theorems; in the algebraic-geometric theory, by the residue

¹The development of the theory of algebraic functions in earlier and more recent times. *Jahresbericht der Deutschen Mathematiker-Vereinigung* vol. 3 (1894) [cited as Brill–Noether].

²With the exception of a discussion of Kronecker’s investigation of the discriminant. (*J. f. M.* vol. 91.)

³The parts concerning Dedekind–Weber are due to A. Ostrowski.

⁴The theory of algebraic number fields. *Jahresbericht der Deutschen Mathematiker-Vereinigung* vol. 4 (1894/95).

⁵Braunschweig, Vieweg, 4th ed. 1894.

theorem. In particular, the zeros of the integrands of the first kind correspond to the differential class in the arithmetic theory, and to the doctrine of special groups (that is, the pencil of point groups cut out by the adjoint curves of order $(n - 3)$, the φ -curves) in the algebraic-geometric theory.

Just as Riemann's theory historically preceded the other theories, even though the rigorous proof of the existence theorems was obtained only later, so also later, with its transcendental aids, algebraic questions were settled which to this day are not yet accessible to a purely algebraic or arithmetic treatment; here above all Hurwitz's theory of singular correspondences is to be mentioned.⁶

Analogies with the transcendental questions in the theory of algebraic number fields will be discussed in the last section.

For the remaining theories the algebraic functions are primary, the integrals secondary. In this respect the Weierstrass theory rests on the transcendental foundations of power-series expansion and the residue theorem; but it then proceeds algebraically, by explicit specification of all functions under consideration.

The arithmetic theory in the form of Hensel–Landsberg also uses the transcendental foundations of series expansion. But by assigning divisors to the individual points on this foundation, it then proceeds purely arithmetically and can thus be viewed as a fusion of Weierstrass and Dedekind–Weber. In contrast with the latter, it simultaneously supplies methods for the actual construction of all basis functions in question. In detail it exhibits a number of simplifications compared with Dedekind–Weber, essentially by the introduction of “fractional” divisors. Recently Hensel⁷ has given an arithmetic foundation for the theory of series expansions (apart from a majorant method for the convergence proof), whereby the theory has become almost purely arithmetic.

The algebraic-geometric theory of Brill–Noether presupposes only the concept of “successive points” of a branch, which is reduced to the series expansion at an ordinary point⁸ and corresponds to the Weierstrass element concept; it then continues with algebraic-rational methods, essentially those of ternary elimination theory; it too reaches explicit representations.

The original, purely arithmetic theory of Dedekind–Weber⁹ is akin to Riemann's in that its theorems again have the character of existence proofs; it preserves complete purity of method.¹⁰ The arithmetic foundation of the theory, where the variables play only the role of indeterminates, runs parallel to the theory of algebraic numbers, as will be explained more closely in 2.¹¹ This foundation gives the means of introducing the point concept arithmetically (compare Hensel no. 10a) and thus of setting up the concept of the absolute Riemann surface without any consideration of continuity. The theory thereby acquires a formal character; for example, the concept of the differential is also introduced purely formally. On the basis obtained by the point concept, the close kinship with the methods and concepts of the algebraic-geometric theory can be demonstrated, since the latter too, apart from the element concept, is free of transcendental aids; this is to be done in the following sections.

2. Parallelism between algebraic numbers and algebraic functions. (Conjugate elements, basis theorems, module concept)

An algebraic number field is defined as a field of degree n over the field R of rational numbers; a field of algebraic functions of one variable as a field of degree n over the field $K(z)$ of all rational functions of one indeterminate z with arbitrary complex coefficients. Thus for both fields all theorems hold in common which abstract completely from the special nature of the ground field; these are the theorems on bases, norms, traces, and discriminants.¹² All these concepts are introduced by Dedekind–Weber without the use of conjugate elements. It should be noted, however, that the concept of the conjugate field can be defined in an entirely formal way, as Steinitz showed following Kronecker.¹³ Namely, if K is an arbitrary field and $\varphi(x)$ a polynomial in x irreducible over K , then the system of residue classes modulo $\varphi(x)$ forms a field. If one assigns to the residue class of x an element ξ , then under the isomorphic assignment the element $f(\xi)$

⁶A. Hurwitz, Über algebraische Korrespondenzen und das erweiterte Korrespondenzprinzip. *Math. Ann.* vol. 28. (Compare Brill–Noether, section 10.)

⁷*Mathematische Zeitschrift* vol. 4; the foundation is also used as the basis of his report.

⁸For successive “singular” points compare Brill–Noether, section 6, especially nos. 10 and 11. Ordinary successive points are, for example, the points of intersection of the tangent with the curve or higher points of contact (branch points).

⁹*J. f. M.* 92 (cited as D.–W.).

¹⁰For this reason it is also more suitable for comparison with the other theories than, for instance, the theory of Hensel–Landsberg, which will occur only occasionally in the following comparisons. — The theory of Abelian integrals and their inversion, insofar as it presupposes continuity, is however not treated in D.–W. nor in the further development of the algebraic theory (M. Noether, *Ann.* 37).

¹¹A comparison of the theory of Hensel–Landsberg with number theory, especially the theory of p -adic numbers, is found in Hensel.

¹²D.–W., § 1 and 2; Hensel no. 4 (with note 5).

¹³Steinitz, *Algebraische Theorie der Körper.* (*J. f. M.* 137, § 6 and 8.)

corresponds to the residue class represented by the polynomial $f(x)$. Since all polynomials divisible by $\varphi(x)$ represent the zero class, $\varphi(\xi)$ is assigned to the zero class; the element ξ can thus symbolically be regarded as a zero of $\varphi(x) = 0$ and generates the algebraic field $K(\xi)$. Repetition of the procedure leads to the n conjugate fields $K(\xi), K(\xi'), \dots, K(\xi^{(n-1)})$. Thus one may in particular speak of conjugate elements also for algebraic functions, without leaving the purely arithmetic theory.

The two ground fields R and $K(z)$ have the property in common that in them the system of all integral quantities is defined, as rational integers and as integral rational functions of one indeterminate, respectively. These integral quantities have the characteristic property that any two a and b possess a greatest common divisor d that is representable in the form $ma + nb$, where d, m, n are also integral quantities from R or $K(z)$; and indeed d is the absolutely smallest number, respectively the polynomial of lowest degree, representable in this form. On this property alone, which entails the unique decomposition into prime factors for rational integers and for functions of one indeterminate, rest the theorems on the integral quantities of the algebraic or upper field. The argument proving the existence of the greatest common divisor also gives the existence of a fundamental system or a basis of the integral quantities of the algebraic field. If one considers only such bases of the field as consist of integral quantities, then their discriminant becomes a rational integer, respectively a rational function of one indeterminate. Every basis that corresponds to the absolutely smallest number, respectively to the function of lowest degree, forms a fundamental system.

A more precise insight into the basis questions, also for more general fields, is given by the introduction of the module concept, which shall be formulated so as to include the definitions that occur differently in the literature according to the nature of the elements. A module — with respect to a basic domain of integrity $\mathfrak{S}$ — is a system of elements such that the difference of two elements again belongs to the system, and likewise the product of an element with an arbitrary quantity from $\mathfrak{S}$. (If $\mathfrak{S}$ consists of the rational integers, then the second requirement is a consequence of the first.)¹⁴

Every module that has a basis of finitely many elements through which every element can be represented linearly with coefficients from $\mathfrak{S}$ is called finite (finitely generated); in particular, every one-term module is therefore of the form $c \cdot \alpha$, where α denotes the basis element and c runs through all quantities from $\mathfrak{S}$. A module whose elements themselves belong to $\mathfrak{S}$ is called an ideal in $\mathfrak{S}$; whence follows the concept of a finite ideal, and in particular of a one-term ideal (principal ideal).

Now all basis theorems for integral quantities and ideals rest on the following theorem:

*If M is a module with respect to $\mathfrak{S}$, whose elements consist of linear forms in n indeterminates with coefficients from $\mathfrak{S}$, and if every ideal in $\mathfrak{S}$ is finite, then M is also a finite module. If in particular every ideal in $\mathfrak{S}$ is a principal ideal, then M possesses a basis of linearly independent forms.*¹⁵

Indeed, if the elements of M are given by $l(x) = a_1x_1 + \dots + a_nx_n$, then the totality of the coefficients a_i runs through an ideal in $\mathfrak{S}$, which by hypothesis is of the form $b_1a^{(1)} + \dots + b_\rho a^{(\rho)}$. The linear form belonging to M ,

$$m(x) = l(x) - b_1l^{(1)}(x) - \dots - b_\rho l^{(\rho)}(x),$$

therefore depends only on $(n-1)$ indeterminates, from which the assertion follows by finitely many repetitions. If in each case $\rho = 1$, then the basis consists of n forms in respectively $n, n-1, \dots, 1$ indeterminates, which gives the linear independence of the non-vanishing forms.

The theorem on the fundamental system follows from this in the various cases as follows:

If the algebraic field is obtained by adjoining the integral algebraic element δ to the ground field, and if $\mathfrak{S}$ denotes the totality of all integral quantities of the ground field, then every integral quantity of the upper field admits the representation (compare, for instance, Zahlbericht § 3)

$$\omega = \frac{a_1\delta^{n-1} + a_2\delta^{n-2} + \dots + a_n}{\Delta(\delta)},$$

where the a_i are quantities from $\mathfrak{S}$ and $\Delta(\delta)$ denotes the discriminant of δ . By the substitution

$$x_i = \frac{\delta^{n-i}}{\Delta(\delta)}$$

¹⁴The module concept is due to Dedekind, who first introduced the number module ($\mathfrak{S}$ = rational integers) in the second edition of the number theory; the module of integral linear forms is also implicitly contained there. In Dedekind-Weber the “function modules” occur, whose elements consist of algebraic functions, whereas $\mathfrak{S}$ consists of all integral rational functions of one indeterminate. — The module of homogeneous forms in n indeterminates, more generally of polynomials, occurs first in Hilbert (*Annalen* 36); $\mathfrak{S}$ consists of all polynomials, and since the elements are also polynomials, this module falls under our ideal definition. Kronecker’s “module systems” of polynomials start from the basis, not from the abstract definition.

¹⁵The theorem appears in the literature separately for the different coefficient domains $\mathfrak{S}$. For rational integers compare, for instance, Hilbert, Zahlbericht § 3 and 4; for polynomials in several indeterminates: Hilbert, Über die vollen Invariantensysteme: end of § 2 (*Math. Ann.* 42).

the module of all integral quantities is transformed into a module of linear forms; hence the existence of the fundamental system follows in all cases where every ideal in $\mathfrak{S}$ is finite. The same argument shows more generally that every ideal in the domain of integral quantities of the upper field is finite. Now in particular for rational integers and for functions of one indeterminate every ideal is a principal ideal; hence it follows that the fundamental systems in algebraic number fields and in fields of algebraic functions of one indeterminate consist of exactly n quantities, when n denotes the degree. And since, in these fields, every ideal is finite by the preceding argument, this at the same time yields the fundamental system for relative fields, which in general will consist of more quantities than the relative degree indicates.¹⁶ Further, if $\mathfrak{S}$ denotes the domain of all polynomials in several indeterminates, every ideal in $\mathfrak{S}$ is also finite;¹⁷ hence the existence of the fundamental system is obtained also for algebraic function fields of several indeterminates, and it will again consist of more quantities than the degree indicates.

3. Parallelism and differences in the ideal theory of algebraic numbers and functions

The proof of the fundamental theorem of ideal theory, that every ideal can be represented uniquely as a product of powers of prime ideals, can be carried out jointly for algebraic numbers and functions; solely on the basis of the property that, for rational integers and for functions of one indeterminate, every ideal is a principal ideal.¹⁸ The main point of the proof lies in showing that, from the divisibility of an ideal $\mathfrak{c}$ by an ideal $\mathfrak{a}$ (that is, every element of $\mathfrak{c}$ is contained in $\mathfrak{a}$), the product representation $\mathfrak{c} = \mathfrak{a}\mathfrak{b}$ follows.¹⁹ In Hurwitz this proof rests on the “extended Gauss theorem”, which says that an integral quantity ω which divides every coefficient of the product of two polynomials φ and ψ also divides the product of all coefficients of φ and ψ ; in Dedekind it rests on a module theorem equivalent to this theorem.²⁰

Going further, the concept of the complementary module (Hensel no. 12), and from it the ramification ideal (ground ideal, different) $\mathfrak{D}$, can be defined jointly (its norm becomes the field discriminant D); and the fundamental theorem holds that the ramification ideal $\mathfrak{D}$ is the *greatest common divisor of all principal ideals* $F'(\delta)$; here δ runs through all integral quantities of the field and $F(\delta) = 0$ denotes in each case the corresponding irreducible equation. From this it follows that the *field discriminant contains all and only those rational primes, respectively linear factors in z , which are divisible by the square of a prime ideal*. For $F'(\delta)$ itself one obtains the representation

$$F'(\delta) = \mathfrak{f} \cdot \mathfrak{D},$$

and by taking norms

$$\Delta(\delta) = R^2 \cdot D;$$

here $\mathfrak{f}$ denotes the conductor of the ring derived from δ (order, domain of integrity), that is, $\mathfrak{f}$ consists of the totality of the (integral) quantities which, after multiplication by any integral quantity, are divisible by the module $[1, \delta, \dots, \delta^{n-1}]$. $R^2 = \text{Norm } \mathfrak{f}$ represents the square of the determinant of substitution that carries the basis $1, \delta, \dots, \delta^{n-1}$ into a basis of the integral quantities.²¹

Differences in the further development are conditioned by special properties of the ground fields. Thus, in the case of algebraic numbers, the fact that the *elements of the ground field are at the same time exponents* leads to the possibility that the ramification ideal may be divisible by a higher power than the $(e-1)$ -st power of a prime ideal $\mathfrak{p}$ which occurs in p to the e -th power (namely when e is divisible by p); and this leads to the further concepts of the inertia field and ramification field,²² which have no analogue in the theory of algebraic

¹⁶According to Steinitz's investigations on modules of linear forms, where $\mathfrak{S}$ consists of the integers of a (finite) algebraic number field, $\nu+1$ basis elements suffice when ν denotes the relative degree (*Math. Ann.* 71, § 4).

¹⁷By Hilbert's theorem on the module basis (*Math. Ann.* 36). The modules of polynomials here called ideals for the sake of consistency are usually called “form modules” or simply “modules”. Hilbert's theorem that such a module N is finite can also be reduced to the theorem on linear forms. Namely, let f be a polynomial of degree n contained in N in the $r+1$ indeterminates $x_1, \dots, x_r, x_{r+1}$, containing the term x_{r+1}^n (which can always be achieved by a linear transformation); then all polynomials from N are congruent mod. f to those representable in the form $a_1(x)x_{r+1}^{n-1} + \dots + a_n(x)$; hence, by $x_{r+1}^{n-i} = X_i$, they form a module of linear forms for which every ideal in the domain of r indeterminates may be assumed finite, since this is the case for one indeterminate.

¹⁸Compare a remark in the introduction to Hurwitz: *Über die Theorie der Ideale* (*Gött. Nachr.* 1894). In Weber's textbook of algebra the proof, following Kronecker, is carried out in parallel for both fields by means of the introduction of functionals. [The connection between Kronecker's theory and ideal theory is mediated by the “extended Gauss theorem”; Hurwitz, loc. cit.]

¹⁹Emphasized especially by Dedekind (number theory); Supplement XI, theorem VII, § 177; compare the note on p. 554. The theorem uses the fact that algebraic fields are involved; it no longer holds for ideals (modules) of polynomials, where the least common multiple takes the place of the product.

²⁰Number theory, Suppl. XI; theorem VI, § 173. Dedekind points out the equivalence of the two theorems (*Über die Begründung der Idealtheorie*, *Göttinger Nachr.* 1895).

²¹Dedekind, *Über die Diskriminante endlicher Körper* (*Abhandl. der Gött. Ges. d. Wissensch.* vol. 29, 1882). The definitions given there for algebraic numbers can be transferred directly to algebraic functions; the proofs too can be transferred almost exactly. The order of proof in D.-W. is different: see below. For the cited theorems compare also *Zahlbericht*, § 31 and 32. ($F'(\delta)$ is there called the “different” of the integer δ .)

²²Compare *Zahlbericht*, § 39–45.

functions. Above all, however, because of the *existence of infinitely many constants* in the ground field, the finiteness of the number of ideal classes disappears for algebraic functions, and with it all questions connected with determining the class number. Nevertheless, in a certain sense the domain of all (transcendental) prime forms²³ may be regarded as a transcendental analogue of the class field; for here every point is generated by a single form, a principal ideal.

On the other hand, the existence of infinitely many constants in the ground field $K(z)$ leads, for algebraic functions, to theorems and questions which have no analogue for algebraic numbers. First, a certain simplification in the arrangement of the proof follows from this; for example in the proof given by D.–W. of the unique factorization of ideals into prime ideals, where the “extended Gauss theorem” or a theorem equivalent to it can be avoided by using the fact that every linear form $(z - c)$ has infinitely many residue classes, namely all constants (whereas the number of residue classes modulo a prime number is finite).²⁴

The further fact that every polynomial with integral rational functions of z as coefficients decomposes into linear factors mod. $(z - \alpha)$ (which is not the case for an integral polynomial mod. p) leads to genuinely new results; this fact also still holds if, instead of all constants, one takes only all algebraic numbers.²⁵ From this it follows, namely, that every prime ideal is an ideal of first degree;²⁶ and further, that the field discriminant D becomes the intersection of all discriminants $\Delta(\delta)$, where δ runs through all integral quantities;²⁷ both are in contrast to algebraic numbers. Subsequently it again follows from this that the ramification ideal is the intersection of all principal ideals $F'(\delta)$.

Further differences are conditioned by the fact that, for algebraic numbers, the ground field R of rational numbers is something absolutely distinguished, namely the prime field contained in the field (that is, the field derived from the unit); whereas the ground field $K(z)$ is something relative. Every nonconstant function ξ may be chosen as independent variable; the field then becomes an algebraic field over $K(\xi)$. The theorem that every prime ideal $\mathfrak{p}$ is an ideal of first degree, hence every function is congruent modulo $\mathfrak{p}$ to a constant, combined with the possibility of taking an arbitrary function as independent variable, leads to the most important concept going beyond algebraic numbers, the point concept in purely arithmetic form. (Compare Hensel 10a.) “Every point is represented here by a field of constants isomorphic to the function field.”²⁸ The totality of points so defined forms the “absolute Riemann surface”. This definition of point depends only on the field, not on a special variable. The existence proof, however, is carried out by distinguishing some function as independent variable z ; a point P is generated by a prime ideal $\mathfrak{p}$ by assigning to every function its constant residue modulo $\mathfrak{p}$; thus in particular the functions divisible by $\mathfrak{p}$ vanish at P , and conversely. As $\mathfrak{p}$ runs through all prime ideals in z , all points in which z is finite are thereby obtained; the remaining points arise by introducing the independent variable $\xi = 1/z$ and correspond to the prime ideals which divide the principal ideal ξ . Naturally, the concepts attached to the point concept also can have no analogue for algebraic numbers; such as the concept of a polygon, of a polygon class, of the dimension of a polygon class, and so on.

It should also be noted that the free choice of the independent variable leads to a further interpretation of the decomposition $F'(\vartheta) = \mathfrak{f} \cdot \mathfrak{D}$. Namely, if one considers ϑ as independent variable, then correspondingly one obtains

$$\left(\frac{\partial F}{\partial z}\right) = \mathfrak{f} \cdot \mathfrak{a},$$

since the ring derived from ϑ was defined symmetrically with respect to z and ϑ ; and $\mathfrak{f}$ becomes the greatest common divisor of the two principal ideals $\frac{\partial F}{\partial z}$ and $\frac{\partial F}{\partial \vartheta}$; the conductor $\mathfrak{f}$ becomes at the same time the “double-point ideal” with respect to ϑ, z .²⁹

4. Connection of the arithmetic foundation of series expansion with ideal theory

In Hensel–Landsberg the transcendental aid of series expansion and the point concept borrowed from function theory take the place of ideal theory. Hensel’s recently given arithmetic foundation of series expansion for algebraic numbers and functions³⁰ can therefore be regarded as an equivalent of ideal theory. In fact this

²³Klein, Zur Theorie der Abelschen Funktionen (*Ann.* 36, 1890).

²⁴D.–W. § 9. Compare the notes on pp. 212 and 213.

²⁵D.–W. note in the introduction that in this case the whole theory remains intact.

²⁶D.–W., § 9, no. 7.

²⁷D.–W., p. 224.

²⁸Two fields are, as is well known, called “isomorphic” if they can be put in one-to-one correspondence in such a way that sum, difference, product, and quotient also correspond.

²⁹D.–W. p. 263.

³⁰Eine neue Theorie der algebraischen Zahlen (*Math. Zeitschr.* vol. 2, 1918) and Neue Begründung der arithmetischen Theorie der algebraischen Funktionen einer Variablen (*Math. Zeitschr.* vol. 4, 1919). For a more detailed exposition compare Hensel no. 44ff.

arithmetic foundation amounts to passing, for each prime ideal, to such a (transcendental) extension field in which this prime ideal becomes a principal ideal, so that the usual decomposition theorems hold; and it suffices to carry out this consideration for all prime ideals occurring in one prime number p , respectively one linear function $z - a$.

First one introduces a transcendental extension field formed by all series expansions in integral powers of $z - a$, respectively p (p -adic numbers), with in each case at most finitely many negative powers. In this field the irreducible equation defining the algebraic number field or function field decomposes into the product of ρ irreducible factors, corresponding to the decomposition of p , respectively $z - a$, into the product of the ρ “primary” ideals $\mathfrak{q}_1, \dots, \mathfrak{q}_\rho$. The further representation of each primary ideal as a power of a prime ideal,

$$\mathfrak{q}_i = \mathfrak{p}_i^{\ell_i},$$

corresponds in Hensel to the introduction of a further field, algebraic with respect to the transcendental extension field. And in the case of algebraic functions this field can be defined by a pure equation; in the case of algebraic numbers, more generally, by an algebraically solvable equation. The simplification for algebraic functions is due to the fact that here no inertia and ramification fields (compare no. 3) occur.

The “terminating series” in a prime element π^{31} that occurs in these investigations (and which does not yet require a transcendental convergence proof)

$$v = c_0 + c_1\pi + \dots + c_{\sigma-1}\pi^{\sigma-1} - v_\sigma\pi^\sigma,$$

where $c_0, \dots, c_{\sigma-1}$ are constants and v_σ denotes an integral element of the field, is found in exactly corresponding form in D.-W.³² There it is derived from ideal theory and gives the basis for defining the order of vanishing (respectively becoming infinite) of a function at a point.

5. Comparison of the arithmetic theory of algebraic functions with the geometric theory. Different concept of invariance

“Invariant” means any concept characterized by the field alone. In the arithmetic theory this invariance is in general already given in the definition, which is stated with respect to all elements of the field, not with respect to some basis or distinguished variable; the point concept given above is an example. By contrast, the remaining theories start from some special variable or basis and show afterwards the independence of the concept from this special choice; invariance here becomes invariance under birational transformation.

Namely, suppose that the field arises on the one hand by adjoining the algebraic element s to $K(z)$, and on the other by adjoining σ to $K(\xi)$, where $f(z, s) = 0$, respectively $F(\xi, \sigma) = 0$, denote the irreducible equations for s , respectively σ ; or, still more generally, by adjoining a finite number of elements to $K(\eta)$, with the corresponding irreducible equations

$$g_1(\eta, \tau_1) = 0, \quad g_2(\eta, \tau_1, \tau_2) = 0, \dots$$

Then, by definition, all elements can be expressed rationally by z and s , just as by ξ and σ , and also by $\eta, \tau_1, \tau_2, \dots$. In particular z and s are therefore rationally expressible by ξ and σ , and conversely; likewise z and s rationally by $\eta, \tau_1, \tau_2, \dots$, and conversely. The plane curve $f(z, s) = 0$ thus goes by “birational transformation” into the plane curve $F(\xi, \sigma) = 0$, or also into the curve in the space of variables $\eta, \tau_1, \tau_2, \dots$ defined by $g_1(\eta, \tau_1) = 0; g_2(\eta, \tau_1, \tau_2) = 0, \dots$. Invariance therefore appears in the fact that a definition stated for one special curve remains valid for all curves obtained from it by birational transformation (in a space of arbitrarily many dimensions); for all curves of the “class” in Riemann’s sense. A “point” is here defined as a coherent system of values $z = a, s = b$, so that $f(a, b) = 0$; under birational transformation this point generally goes again into a coherent system of values α, β of $F(\xi, \sigma) = 0$, but in special cases, when a, b was a “singular” point, it can correspond to several “points” on $F(\xi, \sigma) = 0$; and one can always specify curves on which a “singular” point of $f(z, s) = 0$ corresponds to a finite number of simple points lying separately or successively (resolution of singularities).³³ Likewise a point group again goes into a point group; and the number of its points is invariant as soon as one counts a singular point as as many points as the simple points corresponding to it under a suitable transformation. Since every curve can be transformed birationally into

³¹Neue Begründung . . . , formula (11).

³²D.-W., p. 231 and § 15.

³³Compare, for instance, Brill–Noether, section 6. More precise information on “singular” points appears in the next section.

one with “ordinary multiple” points,³⁴ the geometric theory develops its concepts only for such special curves and then proves the invariance under every birational transformation, including one which leads to the most general singular points.

In particular, as a distinguished curve of the “class” — excluding the hyperelliptic case — one can introduce the so-called “normal curve of the φ ’s” in the space of $(p - 1)$ dimensions, which has no singular points; here the φ ’s are the p “adjoint curves of order $(n - 3)$ ” for the homogenized $f(z, s) = 0$. A birational transformation of $f(z, s) = 0$ into $F(\xi, \sigma) = 0$ corresponds to a linear transformation of the φ ’s, so that this is invariance in the projective sense. The representation of all functions of the field as rational functions of the φ ’s is called the “invariant representation” in the geometric theory.³⁵

The linear system of the φ ’s, or also of the “special groups” cut out by them, therefore goes into itself under every birational transformation, and is consequently characterized by the field alone. It corresponds to the differential class in the arithmetic theory, where the invariance is again shown directly in the definition, which is given by properties with respect to every function of the field as independent variable.

6. Continuation of the comparison. Singular points

The different conception of the point concept is connected with the fact that the “singular points” of curves, which cause special difficulties in the algebraic-geometric theory, do not occur at all in the foundations of the arithmetic theory; and consequently the “adjoint functions” are not required for the construction of the arithmetic theory, as they are in the geometric theory, but belong to special higher parts of the theory. On the other hand, the branch points do not occur in the foundations of the geometric theory; in fact, they also drop out in the representation of the integrals in Aronhold’s normal form (Hensel no. 27, formula 80).

The definition of singular point given in 5 can be formulated as follows: $f(z, s) = 0$ (respectively a curve in a higher-dimensional space) has a singular point at $z = a, s = b$ (respectively $z = a; s_i = b_i$) if this system of values is assumed by more than one point of the absolute Riemann surface (point in the arithmetic sense), or if it is assumed at one or several points with higher order. In the latter case one has a point of the ramification polygon of z as well as of s ; this corresponds to the “successive points” of the geometric theory; the point becomes a (higher) cusp point of $f(z, s) = 0$ (respectively of the curve in the higher-dimensional space).

Since by hypothesis every arbitrary function η of the field can be represented rationally by z and s , because of the isomorphism every point of the absolute Riemann surface at which $z = a, s = b$ is obtained by assigning to every function η , represented by z and s , its value for $z = a, s = b$. Thus, for the point to be singular, there must be at least one function η which assumes several values or one system of values several times for $z = a, s = b$. And because of rational representability by z and s , this can be the case only when in this representation numerator and denominator vanish simultaneously for $z = a, s = b$. It follows that the above definition is identical with the usual one, namely that

$$f, \quad \frac{\partial f}{\partial z}, \quad \frac{\partial f}{\partial s}$$

vanish for $z = a, s = b$. For in this case

$$\eta = \frac{\frac{\partial f}{\partial z}}{\frac{\partial f}{\partial s}}$$

is a function of the indicated kind which is multivalued for $z = a, s = b$, whereas in the opposite case every function, even if it becomes $0/0$, assumes only one value.

With the arithmetic point concept the singular point cannot occur, since two points are regarded as different as soon as only one function is assigned different systems of values. But even in the ideal theory serving to found the arithmetic point concept, the concept of a singular point does not occur, precisely because the totality of all integral quantities of the field is always considered at the same time. All singular points in the finite part of $F(z, \delta) = 0$, where δ is algebraically integral with respect to z , are generated (according to the second definition and the end of no. 3) by the “double-point ideal” $\mathfrak{f}$. Now, since by the fundamental theorem the ramification ideal $\mathfrak{D}$ is the greatest common divisor of all principal ideals $F'(\delta)$, for every prime ideal $\mathfrak{p}$ one can specify a δ such that the corresponding $\mathfrak{f}$ is not divisible by $\mathfrak{p}$, and consequently — since $\mathfrak{f}$ is the greatest common divisor of $F'(\delta)$ and $F'(z)$ — at least one of the two principal ideals $F'(\delta)$ and $F'(z)$ is not divisible by $\mathfrak{p}$. Thus there is *no system of values for which all $F'(\delta)$ and $F'(z)$ vanish simultaneously*, and

³⁴Ibid.

³⁵Compare, for instance, Brill-Noether, section 8. For algebraic functions of several variables the question whether invariance can be reduced to such invariance under linear transformation is not treated.

consequently the system of values $z = a, \delta_i = b_i$ is assumed at only one point of the absolute Riemann surface; the *totality of integral algebraic functions of the field* (which may be interpreted as a curve in a space of infinitely many dimensions) *has no singular point in the finite part*; only finite values enter into consideration in ideal theory.

But the basis functions $\omega_1, \dots, \omega_n$ of all integral quantities also have no singular points in the finite part. By the preceding, each point in the arithmetic sense is already uniquely determined by a system of constants isomorphically assigned to all integral functions. If, therefore, a “space curve” defined by any integral functions s_i has a singular point in the finite part for $s_i = \alpha_i$, then there must be at least one integral function η which assumes several systems of values for $s_i = \alpha_i$. If now $\omega_1, \dots, \omega_n$ denotes a basis of all integral quantities, then, because $\delta = a_1\omega_1 + \dots + a_n\omega_n$ with integral rational a_i , each system of values of the ω ’s corresponds to only one system of values of δ ; hence the space curve defined by $\omega_1, \dots, \omega_n$ can possess no singular point in the finite part; and *every point of the absolute Riemann surface at which z is finite is already uniquely defined by a system of constants isomorphically assigned to the ω ’s. The basis functions are precisely the functions η which become multivalued for $s_i = \alpha_i$* . Thus by introducing the basis of integral quantities, for example in place of a basis $1, \delta, \dots, \delta^{n-1}$, the difficulty of singular points is avoided.

In the geometric theory the task of generating all points of the absolute Riemann surface uniquely — more generally, of forming all functions which become infinite at given points — is accomplished by means of adjoint forms and their quotients. A form ψ is called adjoint if $\psi = 0$ has at least an $(i-1)$ -fold point at each i -fold point of the base curve $f(z, s) = 0$; a higher singular point must first be resolved into ordinary multiple points. It was shown above that, for the most general function $\eta = \frac{\psi}{\chi}$, numerator and denominator must vanish at all singular points of $f(z, s) = 0$; that they must be adjoint is shown by considering the everywhere finite differentials. That, conversely, the conditions of adjointness are also sufficient for forming the most general function is shown by the “residue theorem”, which will be discussed more closely below. Thus the adjoint functions take the place of the basis functions of the arithmetic theory.

Arithmetically, an adjoint form corresponds to the numerator of a function of the field divisible by the “double-point ideal” $\mathfrak{f}$, as soon as one assumes the singular points — which can always be achieved by transformation — all to lie in the finite part. From this the representation of the basis quantities $\omega_1, \dots, \omega_n$ as quotients of adjoint forms, and hence the connection among the various ways of representing them, is shown formally as follows:

Let $R(z)$ denote the determinant of substitution of a basis $1, \delta, \dots, \delta^{n-1}$ into a basis $\omega_1, \dots, \omega_n$. Conversely, every ω can then be expressed as a quotient of an integral function of z and δ by $R(z)$:

$$\omega_i = \frac{G_i(z, \delta)}{R(z)}.$$

Since $R(z)\omega_i$ belongs to the ring derived from δ , by the definition of the conductor $R(z)$ is divisible by $\mathfrak{f}$ [from $(R(z))^2 = \text{Norm } \mathfrak{f}$ this divisibility follows only for $(R(z))^2$]. Since now ω_i , as an integral function, remains finite for all finite z , the numerator function $G_i(z, \delta)$ must also be divisible by $\mathfrak{f}$; numerator and denominator of all ω_i are therefore adjoint. From this representation of the ω ’s follows, however, the representation of all functions of the field by quotients of adjoints.

According to the preceding, the geometric theory can be characterized as the *theory of the ring derived from a quantity δ (order)*, in contrast with the arithmetic theory, which considers *all integral quantities* simultaneously. Thus it is clear that the conductor of the ring — the singular points — plays a decisive role. Further analogies between the arithmetic theory of rings (orders) and the geometric theory can also be given. Thus the geometric theory does not count the intersection points that fall into the singular points in the intersection with adjoint curves, but considers only the point groups different from these. Corresponding to this is that Dedekind³⁶ considers only such ring ideals as are prime to the conductor ideal $\mathfrak{f}$, and for this domain of ideals sets up all divisibility laws, including unique decomposition into prime ideals.

Also the “fundamental theorem of algebraic functions”³⁷ underlying the geometric theory, or at least an immediate consequence of it, has in a certain sense an analogue in the arithmetic theory of rings. This theorem gives the conditions for a representability

$$\psi(z, s) = A(z, s)f(z, s) + B(z, s)\varphi(z, s),$$

where all quantities occurring are integral rational in s, z (more generally integral and homogeneous in

³⁶Über die Anzahl der Idealklassen in den verschiedenen Ordnungen eines endlichen Körpers. (Festschrift Braunschweig 1877); § 3-5. (The theorems stated in these paragraphs for algebraic numbers hold literally for algebraic functions of one variable.)

³⁷M. Noether, Über einen Satz aus der Theorie der algebraischen Funktionen. *Math. Ann.* vol. 6 (1873).

x_1, x_2, x_3). From these conditions it follows,³⁸ by virtue of $f(z, s) = 0$, that the quotient $\frac{\psi}{\varphi}$ is always equal to an integral rational function $B(z, s)$ when it is adjoint to f . Corresponding to this is the theorem from no. 3, which underlies all these comparisons, that the conductor $\mathfrak{f}$ of a ring derived from δ is equal to the double-point ideal of $f(z, \delta) = 0$. For, by the definition of the conductor, every algebraic integral quantity divisible by $\mathfrak{f}$ belongs to the ring, and is therefore integral and rationally representable by z and δ ; whereas the cited theorem says that such a quantity is an adjoint function.³⁹

7. Comparison between the residue theorem and theorems of ideal theory

The just mentioned consequence of the “fundamental theorem of algebraic functions” leads in the geometric theory directly to the “residue theorem”, and thus to the foundation on which the whole geometric theory is built. In a certain respect, however, the residue theorem again has a more elementary character than the remaining foundations, insofar as it can be stated arithmetically as a direct consequence of the unique factorization of ideals into prime ideals, or rather of the facts underlying this decomposition theorem, and therefore does not rest on the higher theory of rings. One is led to this arithmetic formulation as follows:

Assume the base curve $f(z, \delta) = 0$ — as can always be achieved by a linear fractional transformation of the variables — to be such that the singular points and the finitely many point groups under consideration lie in the finite part; and further that δ depends algebraically integrally on z , which can always be achieved by a subsequent linear transformation integral in the variables. Then to a prime ideal $\mathfrak{p}$ generating a point P of the absolute Riemann surface at which $z = a, \delta = b$, there corresponds the point $z = a, \delta = b$ of $f(z, \delta) = 0$; more generally, an ideal $\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_\rho^{e_\rho}$ generates the point group consisting of the points $z = a_i, \delta = b_i$ with the corresponding multiplicity; and all points of $f(z, \delta) = 0$ lying in the finite part are so generated. Since at a point P all functions $g_1, g_2, \dots$ divisible by $\mathfrak{p}$ vanish, and since conversely this totality of functions g generates the ideal $\mathfrak{p}$, the corresponding point on f is the intersection of

$$g_1(z, \delta) = 0, \quad g_2(z, \delta) = 0, \dots$$

with $f = 0$; the same holds for the point group corresponding to an ideal $\mathfrak{a}$. Indeed, by the theorem that every ideal is the greatest common divisor of two principal ideals, two such curves always suffice to cut out a point group. In particular, the point group corresponding to a principal ideal is cut out by a curve $g(z, \delta) = 0$ and conversely; the principal ideal corresponds to the complete intersection.

Two point groups A and B are called *corresidual* in the geometric theory if they can be completed with the same point group R to a full intersection. If these point groups are generated respectively by the ideals $\mathfrak{a}, \mathfrak{b}, \mathfrak{r}$, then between these ideals one has the relation

$$\mathfrak{a} \cdot \mathfrak{r} = (\alpha), \quad \mathfrak{b} \cdot \mathfrak{r} = (\beta),$$

which shows that the ideals $\mathfrak{a}$ and $\mathfrak{b}$ are equivalent. Conversely, from the equivalence of two ideals, $\mathfrak{a} = \eta \mathfrak{b}$, this relation always follows on the basis of the theorem that every ideal can be transformed into a principal ideal by multiplication by another ideal;⁴⁰ *equivalent ideals and corresidual point groups are therefore identical concepts*.

The residue theorem now says: *If A and B are corresidual, then they can be cut out by adjoint curves with the same residue (outside the singular points)*. Since the zeros and poles of a function are always corresidual, this proves at the same time that the most general functions of the field are representable as quotients of adjoint forms. In the language of ideal theory the residue theorem simply amounts to saying that, besides $\mathfrak{a}$ and $\mathfrak{b}$, $\mathfrak{a}\mathfrak{f}$ and $\mathfrak{b}\mathfrak{f}$ are also equivalent, where $\mathfrak{f}$ denotes the double-point ideal. For then, by the above, there always exists an ideal $\mathfrak{m}$ such that

$$\mathfrak{a}\mathfrak{f}\mathfrak{m} = (\alpha), \quad \mathfrak{b}\mathfrak{f}\mathfrak{m} = (\beta)$$

become principal ideals; $\alpha = 0$ and $\beta = 0$ are the adjoint curves which cut out the point groups A and B , respectively their subgroups A' and B' different from the singular points.⁴¹ The proof of the residue theorem

³⁸Brill–Noether, no. 55.

³⁹Following Dedekind, the assumption is made here throughout that δ is an algebraic integer (which can always be achieved by transformation). The case more exactly corresponding to geometry, $f(z, s) = 0$, where s may also depend algebraically fractionally on z , is treated by Hensel–Landsberg; compare Hensel no. 26 and 29.

⁴⁰Compare Dedekind, number theory § 181; namely, if $\mathfrak{r}$ is chosen so that $\mathfrak{a}\mathfrak{r}$ becomes a principal ideal (α) , then $\mathfrak{b}\mathfrak{r} = \eta^{-1} \cdot (\alpha)$; and since $\mathfrak{b}\mathfrak{r}$ contains only integral quantities, the same holds for $\eta^{-1} \cdot (\alpha)$, which is therefore also a principal ideal (β) .

⁴¹If $\mathfrak{a} = \mathfrak{f}\mathfrak{a}_1$, $\mathfrak{b} = \mathfrak{f}\mathfrak{b}_1$, $\mathfrak{r} = \mathfrak{f}\mathfrak{r}_1$, where $\mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{r}_1$ are relatively prime, then already $\mathfrak{a}_1\mathfrak{f}\mathfrak{n}$, $\mathfrak{b}_1\mathfrak{f}\mathfrak{n}$ become principal ideals, where $\mathfrak{n}$ is prime to $\mathfrak{f}$. For every ideal can be transformed into a principal ideal by multiplication with an ideal that is prime to a given one (D.–W. p. 214; or number theory, § 178, theorem XI).

is therefore arithmetically contained in the possibility of assigning points to prime ideals, and in the proof that equivalent ideals can be transformed into principal ideals by multiplication by one and the same ideal.

Corresidual point groups need not consist of the same number of points; just as equivalent ideals need not be of the same degree. Thus, for example, all principal ideals are equivalent, and all point groups generated by complete intersections are corresidual. This difference as against equivalent polygons (divisors)⁴² is explained by the fact that those points at which z becomes infinite are not generated by prime ideals in z . In the representation of a function as an ideal quotient $\eta = \frac{\mathfrak{a}}{\mathfrak{b}}$, which states the equivalence of $\mathfrak{a}$ and $\mathfrak{b}$, the zeros or poles of η at which z becomes infinite therefore do not appear. For instance, the principal ideal (α) becomes the product of the prime ideals corresponding to the zeros of the integral function α ; the poles of α do not occur, since α becomes infinite only at points where z becomes infinite. In this respect ideal theory is the exact analogue of form theory in the geometric theory, where there are likewise only zeros and no poles.

The passage from forms to functions is made geometrically by forming quotients of forms of the same order; arithmetically, this corresponds to the passage from ideal classes to the invariantly defined polygon (divisor) classes. For, since no variable is distinguished here any longer, the representation of a function by polygon quotients gives all zeros and poles. The relation of ideal class (class of corresidual point groups) to polygon class is the following: if $\mathcal{A}$ and $\mathcal{B}$ are the polygons generated by the ideals $\mathfrak{a}$ and $\mathfrak{b}$, and $\mathcal{N}$ is the polygon of the points at which z becomes infinite (the denominator polygon of z and of the integral functions of z), then from the equivalence of $\mathfrak{a}$ and $\mathfrak{b}$ follows the equivalence of $\mathcal{A}$ and $\mathcal{B}$ if and only if $\mathfrak{a}$ and $\mathfrak{b}$ have the same degree; in general only the equivalence of $\mathcal{AN}^\nu$ with $\mathcal{BN}^{\nu'}$ ($\nu \neq \nu'$) follows. The division of ideals (respectively of corresidual point groups) into classes is thus a gathering together of infinitely many polygon classes into one class; it may also be viewed as a division of the polygons into classes modulo a power of $\mathcal{N}$.⁴³

The residue theorem for the case of corresidual point groups consisting of different numbers of points occurs only in purely geometric questions. In the application to algebraic functions, in fact only the case corresponding to polygon classes occurs, that of corresidual point groups of the same number. The “complete linear system” generated by a point group, that is, the totality of point groups corresidual to this point group and of the same number of points, corresponds exactly to the polygon class generated by the corresponding polygon; the concepts “sum of two complete systems” and “product of two polygon classes” also coincide. That such a complete system contains only finitely many linearly independent point groups is a direct consequence of the residue theorem, which reduces the dimension of the system (the number of linearly independent point groups) to the dimension of all adjoint curves of arbitrary but fixed order passing through the residual group. In order to arrive at the actual determination of this dimension, the Riemann–Roch theorem, the geometric theory first derives from the residue theorem the “reduction theorem” (theorem on the fixed points);⁴⁴ and from this the Riemann–Roch theorem.

Quite correspondingly, the arithmetic theory shows that every class of polygons is of finite dimension, and determines its dimension by a special basis of the integral quantities, a normal basis defined by means of the complementary module; thus it obtains the Riemann–Roch theorem as a consequence of the connection of the complementary module with the ramification polygon. In the presentation of D.–W. the reduction theorem also occurs,⁴⁵ though not as a foundation as in the geometric theory. In consequence of the correspondence mentioned in no. 6 between basis and adjoint functions, a certain correspondence of the auxiliary means in the two theories can also be established here. In summary, one may say that the formations of concepts in the two theories essentially coincide, even though, apart from formulation, they differ in their arrangement. The two theories arose independently of one another, the geometric one a decade earlier. It should also be noted that the Riemann–Roch theorem originally, in Riemann and Roch and also in Weierstrass, occurs as a rank theorem; what is at issue is the rank of the matrix $(\varphi_i(x_j))$, where the φ ’s run through the p adjoint curves of order $n - 3$, and the x ’s through the points of the residual group.

8. Transcendental questions for algebraic functions and algebraic numbers

Hilbert in particular pointed out analogies between transcendental questions for algebraic functions and for algebraic numbers;⁴⁶ these are essentially existence questions. Thus Riemann’s question concerning the

⁴²A polygon A consists of finitely many points of the absolute Riemann surface; two polygons A and B are called equivalent if they are the zeros and poles of a function η ; η then admits the representation by polygon quotients $\eta = \frac{A}{B}$. (D.–W. § 17 and 18.)

⁴³Hensel–Landsberg, 25th lecture, p. 438. — If one represents the two variables z and s by polygon quotients, then this can be understood as binary homogenizing of each of these variables; if one chooses in particular z and s so that they have the same denominator polygon, then the ternary homogenizing customary in geometry arises.

⁴⁴Brill–Noether, no. 58.

⁴⁵D.–W., § 30, no. 2.

⁴⁶Mathematische Probleme, problem 12. (*Göttinger Nachrichten* 1900).

existence of an algebraic function for a given Riemann surface (hence also for given branch points) corresponds to the question of the existence of algebraic number fields with prescribed group and discriminant.⁴⁷ Whereas in the case of algebraic functions the existence proof can be carried out in general, in the domain of algebraic number fields it has been achieved only for special ground fields (rational numbers, quadratic number fields) and only for Abelian groups.⁴⁸ This concerns Kronecker's theorem that every Abelian number field over the rational numbers can be assembled from fields of roots of unity, and the corresponding theorems of Fueter⁴⁹ and Hecke⁵⁰ for relatively Abelian number fields. Thus the desired number fields are actually supplied by a transcendental means, exponential functions, modular functions; in contrast with the pure existence proof for algebraic functions.

The existence proof of algebraic functions is preceded by the existence of everywhere finite integrals. An analogue of this is the existence of the "singular primary numbers" of an algebraic number field defined with respect to a prime number ℓ , whose ℓ -th roots are relatively unramified and supply the first building blocks for the class field.⁵¹ This existence proof is carried out with essential help from the theorem that in the number field there always exist prime ideals with prescribed residue characters; this theorem may therefore be regarded as an analogue of the boundary-value problem on which the existence of the everywhere finite integrals rests.

The everywhere finite integrals provide a transcendental criterion for determining whether two polygons belong to the same class (two point groups of the same number are corresidual), namely by Abel's theorem, which says that the integrals of the first kind, extended over "corresidual paths", vanish; or by the corresponding differential theorem together with its converse.⁵²

As the analogue of Abel's theorem for algebraic number fields one must take the "reciprocity law for ℓ -th power residues", which gives, for the field itself or at least for a suitable upper field, a criterion that two ideals belong to the same class. Its content can also be formulated by saying that in this upper field the singular primary numbers have the same residue character with respect to all ideals of the same class. This latter fact also holds in the field itself, but there it does not coincide with the reciprocity law.⁵³

Finally, Hensel's theory of p -adic numbers supplies the analogue of the power-series expansion of algebraic functions; for this, see Hensel's report.

⁴⁷The treatment of algebraic functions suggested thereby, starting from the group, was carried out by R. König as a special case of more general questions; compare especially: Riemannsche Funktionen- und Differentialsysteme in der Ebene (*J. f. M.* vol. 148) and *Math. Ann.* 78, 79. In place of the algebraic field, König has a special finite module with respect to $K(z)$: there is only "class property".

⁴⁸Only the simple case of the groups occurring for equations of third and fourth degree, and some quite special groups for equations of degree 8, can be handled for arbitrary ground fields: G. Bucht, Über einige algebraische Körper 8. Grades (*Arkiv f. Math., Astron. och Physik*, vol. 6, no. 30.) Compare also E. Noether, Gleichungen mit vorgeschriebener Gruppe and the Seidelmann dissertation cited there. (*Math. Ann.* 78).

⁴⁹*Math. Ann.* 75.

⁵⁰*Math. Ann.* 76.

⁵¹Furtwängler, Reziprozitätsgesetze für Potenzreste mit Primzahlexponenten in algebraischen Zahlkörpern. *Math. Ann.* 67 and 72.

⁵²Compare, for example, the formulation in M. Noether (*Ann.* 37). Since the residue theorem historically grew out of Abel's theorem, the geometric theory speaks of the "sum" of point groups and linear systems, following the integral sums; in contrast to the "product" of the classes in the arithmetic theory.

⁵³Furtwängler, loc. cit.